Arbitraging the Price Space: Funding-Adjusted Cross-Exchange Perpetual Arbitrage

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May 30, 2026

Abstract

This report investigates the viability of systematic cross-exchange arbitrage across cryptocurrency perpetual futures markets. Because raw price spreads are frequently affected by funding rate mechanics and liquidity issues, exploiting transient dislocations requires additional price normalization and microstructure conditions. We implement a price normalization pipeline to isolate pure price deviations from funding accrual effects. To capture mean-reverting trade signals, we apply an Ornstein-Uhlenbeck (OU) process, discretized as an AR(1) model, to generate dynamic, standardized s-scores. To protect against execution slippage and toxic market flow, we introduce Bid-Ask Illiquidity Guards and Cost-Adjusted Expected Value (EV) gates. Out-of-Sample (OOS) empirical validation confirms the persistence of the generated alpha, demonstrating that when protected by regime-specific liquidity filters, cross-exchange perpetual arbitrage remains a viable and robust strategy.

1 Concepts and Methodology

1.1 Arbitrage Concepts

Under the law of one price, perpetual contracts tracking the same underlying assets and governed by identical specifications, especially by the funding mechanisms, should maintain price parity as they represent equivalent economic exposure. This can be interpreted as a no-arbitrage condition, which is guaranteed to arise from frictionless markets.

In practice, however, perpetual prices frequently decouple across exchanges. Arbitrage trades that are supposed to eliminate the deviations tend to have limits involving transaction costs, funding costs, and liquidity risks. Furthermore, perpetual positions are non-fungible and margin-siloed in different exchanges, so they cannot be directly cleared across venues. These real-world inefficiencies and frictions can allow divergent fund flows and local liquidity shocks to drive more mispricing than in perfect markets.

The existence of these practical deviations suggests some potential room for systematic arbitrage strategies to exploit market dynamics and capture the convergence of these perpetuals toward their relative fundamental values. This report focusing on exploring cross-exchange perpetual-to-perpetual arbitrage strategies on the price space. In what follows, we cover our approaches for price normalalization, mean-reversion modeling of price spreads and strategy development.

1.2 Price Normalization for Cross-Exchange Differences

This section defines the normalized price ($P'_{i,t}$), a critical data transformation designed to isolate pure price variance by stripping away the accrual funding effect and standardizing the funding rates across different exchange-specific intervals.

The Accrual Funding Effect

The necessity of price normalization is fundamentally rooted in the mechanics of perpetual swaps. In these markets, the funding rate acts as either an impending liability (a tax) or a guaranteed payments for position holders at the epoch boundary. Driven by market efficiency, these accrued yet unpaid funding payments are continuously priced into the derivative's value. If raw price data is directly fed into a trading algorithm, the price drift caused by this accrual effect can be falsely interpreted as a part of the stochastic market dislocation, generating spurious mean-reversion signals:

- **Positive Funding (Longs Pay Shorts):** The perpetual contract trades at a structural premium to the spot index. This positive rate serves as a direct liability for long position holders. As the epoch settlement approaches, market participants are heavily incentivized to execute the basis trade—shorting the overpriced perpetual contract while buying the spot asset to capture the risk-free funding yield. This resulting deterministic selling pressure, combined with the market continuously pricing the impending liability into the quote, forces the positive basis to converge downward toward the spot anchor.
- **Negative Funding (Shorts Pay Longs):** The perpetual contract trades at a structural discount to the spot index. This negative rate serves as a direct cash dividend to long position holders. As the epoch settlement approaches, market participants are heavily incentivized to execute the basis trade—buying the discounted perpetual contract while shorting the spot asset. This resulting deterministic buying pressure, combined with the market continuously pricing the impending dividend into the quote, forces the negative basis to converge upward toward the spot anchor.

While the mechanisms above describe positive and negative funding as discrete, static regimes for conceptual clarity, empirical market conditions are dynamic. In practice, the normalization formula continuously updates the cumulative net liability, dynamically stripping out the deterministic accrual drift even as the exchange premium oscillates across the zero-bound.

Derivation of the Normalized Price

Several foundational variables are defined to construct this normalization:

- $P_{i,t}$: The original nominal perpetual price on exchange i at time t (e.g., executed trade price or order book quote).
- $P_{index,t}$: The underlying Spot Index price at time t.
- $F_{i,t}^{impl}$: The exchange-reported predicted (implied) funding rate at time t.
- T_i: The total duration of exchange i's funding epoch, expressed in hours (e.g., 8.0 or 1.0).
- Δt_{hr}: The exact elapsed time since the last funding settlement, measured in continuous hours.

1. **Estimated Funding Yield ($\hat{F}_{i,t}^{hr}$)**
   The exchange-reported predicted or most-updated funding rates are normalized to an hourly yield and utilized directly.
   $$\hat{F}_{i,t}^{hr} = \frac{F_{i,t}^{impl}}{T_i}$$
2. **Normalized Price ($P'_{i,t}$)**
   The final normalized price isolates the accrued funding effect by amortizing the expected yield over the elapsed time:
   $$P'_{i,t} = P_{i,t} + \left(P_{index,t} \cdot \hat{F}_{i,t}^{hr} \cdot \Delta t_{hr}\right)$$

   Note: The exchange-published Mark Price ($P_{mark,t}$) can serve as an accurate substitute for $P_{index,t}$ for this approximation as certain exchanges such as Binance use it to determine the funding payments.

Justification and Considerations of the Adjustment

- **The Additive Reversal (+):** The normalization formula *adds* the accrued funding to neutralize the embedded cash flow. In a positive funding regime (where the perpetual trades above spot), the impending funding tax acts as a liability for long positions, structurally suppressing the raw price $P_{i,t}$. Adding the positive accrued funding offsets this liability, lifting the price back to its clean, ex-tax fair value. Conversely, in a negative funding regime (where the perpetual trades below spot), the impending funding payment acts as a guaranteed asset, structurally propping up $P_{i,t}$. Because the accrued funding rate in this regime is a negative value, adding it deducts this embedded funding payment, pulling the raw price down to its true, ex-funding fair value.
- **The Elapsed Time Multiplier (Δt_{hr}):** Multiplying the hourly yield by the exact time elapsed (Δt_{hr}) linearly amortizes the accrued funding. By adding this amortized liability back at every timestamp t, the deterministic time-decay curve is computationally flattened, allowing the subsequent statistical model to trigger solely on orthogonal shocks.
- **Other Funding Components (Interest Rates and Clamping Limits):** Exchange-defined interest rate differentials tend to remain static for sustained periods and clamping effects are inherently already priced into the order book by market participants. For simplicity, we rely on the statistical model to capture these properties characterized in the data.

1.3 Spreads and Mean-Reversion Modeling

Spread Construction ($x_{A,B,t}$)
To execute the strategy across a multi-exchange universe (e.g. $N = 6$), we adopt an exhaustive pairwise architecture. We construct isolated spreads for every possible two-exchange combination, resulting in $N(N-1)/2 = 15$ distinct trading pairs. We denote the spread variables accordingly:

- $P'_{A,t}, P'_{B,t}$: The normalized prices for Exchange A and Exchange B at time t.
- $x_{A,B,t}$: The logarithmic execution spread between the two exchanges.

To standardize variance across assets with drastically different nominal scales (e.g., BTC vs. ETH), we utilize the logarithmic difference. For simplicity, we use a unitary hedge ratio (i.e. $\beta = 1$) to construct the spread, as both contracts represent identical underlying assets with equivalent delta exposure:

$$x_{A,B,t} = \ln(P'_{A,t}) - \ln(P'_{B,t})$$

Mathematically, $\ln(P'_{A,t}) - \ln(P'_{B,t}) \equiv \ln(P'_{A,t}/P'_{B,t})$. For small deviations, this approximates the continuous percentage difference, converting nominal prices into a unitless dimension of relative dislocation.

Ornstein-Uhlenbeck Process and Discretization

Following the statistical arbitrage framework established by Avellaneda and Lee [1], we model the mean-reverting dynamics of each normalized pairwise spread, denoted x_t, using the Ornstein-Uhlenbeck (OU) stochastic process and the discretization of AR(1). The continuous-time OU process provides a rigorous mathematical foundation for characterizing the speed of reversion, the equilibrium mean, and the variance. The dynamics of the spread are governed by the following stochastic differential equation (SDE):

$$dx_t = \kappa(\mu - x_t)dt + \sigma dW_t$$

where:

- $\kappa > 0$ dictates the speed of mean reversion.
- μ represents the long-term dynamic equilibrium mean.
- σ represents the instantaneous volatility of the process.
- dW_t denotes the increment of a standard Wiener process (Brownian motion).

To apply this continuous-time theoretical model to our discrete, high-frequency empirical data (1-minute resolution), the exact solution of the SDE must be discretized. This discretization maps perfectly to an autoregressive model of order 1, AR(1), which is efficiently calibrated via Ordinary Least Squares (OLS) over a rolling lookback window:

$$x_t = a + \phi x_{t-1} + \zeta_t$$

Assuming a fixed time step $\Delta t = 1$ minute, and with ζ_t representing the normally distributed residuals of the OLS regression, the discrete AR(1) coefficients map directly back to the continuous OU domain parameters as follows:

- **Mean Reversion Speed:** $\kappa = -\frac{\ln(\phi)}{\Delta t}$
- **Dynamic Equilibrium Mean:** $\mu = \frac{a}{1-\phi}$
- **Equilibrium Standard Deviation:** $\sigma_{eq} = \frac{\sigma_\zeta}{\sqrt{1-\phi^2}}$

Note: If the AR(1) coefficient $\phi \geq 1$, the estimated mean reversion speed κ becomes undefined or negative. This indicates the spread is non-stationary (exhibiting random walk or explosive behavior), effectively invalidating the arbitrage signal for that specific rolling window.

1.4 Signal Generation with dynamic S-Scores

The base signal relies on the s-score, which standardizes the spread deviation relative to its discrete AR(1) equilibrium. Following the statistical arbitrage framework established by Avellaneda and Lee [1], we define:

$$s_{A,B,t} = \frac{x_{A,B,t} - \mu_{eq,t}}{\sigma_{eq,t}}$$

The sign of the s-score dictates the directional vector required to maintain market neutrality:

- **Short Spread Signal ($s_{A,B,t} > \theta_{entry}$):** Exchange A is trading at an anomalous premium relative to B. The strategy initiates a **Short** position on Perp A and a **Long** position on Perp B.
- **Long Spread Signal ($s_{A,B,t} < -\theta_{entry}$):** Exchange A is trading at an anomalous discount relative to B. The strategy initiates a **Long** position on Perp A and a **Short** position on Perp B.

2 Strategy and Risk Management

2.1 Core Trading Hypothesis

The core theoretical hypothesis posits that while derivative contracts tracking the same underlying asset across exchanges are cointegrated from the perspective of fundamental values, practical anomalies such as market frictions, latency disparities, and localized demand shocks create transient pricing dislocations. While the exact reversion process may not adhere to an Ornstein-Uhlenbeck (OU) process, the AR(1) procedure implied by the OU framework provides robust features for identifying s-score deviations, equilibrium means, and reversal timing. Statistical alpha is systematically extracted by identifying these stochastic deviations, provided that the expected physical reversion magnitude exceeds the aggregated execution costs and microstructure friction.

2.2 Execution Mechanics and Microstructure Guards

To prevent execution on spurious signals or within illiquid order books, the routing pipeline is governed by dynamic pre-trade liquidity filters.

- **Taker-Taker Execution:** To eliminate leg-risk and guarantee simultaneous capture of the dislocation, the baseline model utilizes aggressive Taker-Taker routing. The round-trip transaction cost (C_{trip}) is defined as:

$$C_{trip} = 2(f_{A,taker} + f_{B,taker})$$

  The fees are assumed to be the top tier VIP fees imposed by the exchanges (see he summary on Table 10).

- **Illiquidity Entry Guard:** Before execution, the engine calculates the real-time physical spread of both legs ($\Delta_A + \Delta_B$). If this combined order-book width exceeds a hard threshold (C_{max}), the trade is rejected. Beyond merely bounding immediate slippage costs, this guard serves as a predictive proxy for toxic order flow and adverse selection. In high-frequency microstructure, market makers defensively pull their quotes—drastically widening the spread—when they anticipate informed directional trading or imminent liquidation cascades. By aborting execution when the spread breaches C_{max}, the engine structurally avoids stepping into these toxic directional shocks. Furthermore, in the context of fragmented cryptocurrency venues, abnormally wide or static spreads frequently indicate API desynchronization, matching-engine latency, or "ghost" order books. This guard immunizes the strategy against executing on illusory data anomalies, ensuring capital is only deployed into verifiable, high-confidence liquidity.

- **Cost-Adjusted Expected Value (EV) Gate:** A trade is initiated *only if* the absolute s-score breaches the conviction threshold ($|s| \geq \theta_{entry}$) **AND** the net expected value remains positive. To avoid theoretical mispricing, the Gross Expected Value (Gross EV) is explicitly defined as the directional convergence distance between the real-time executable cross-book spread ($x_{exec,t}$) and the 24-hour trailing moving average of the raw physical mid-spread ($\mu_{raw,24h}$). This expected physical return must exceed the sum of aggregate exchange fees (C_{trip}) and the 24-hour rolling median liquidity toll (τ_{liq}):

$$E[R] = Gross\_EV - (C_{trip} + \tau_{liq}) > 0$$

  Where $Gross\_EV = |x_{exec,t} - \mu_{raw,24h}|$. The strategy only executes when the physical price gap is wide enough to cover all microstructure friction and still yield a net-positive cash flow upon reversion.

2.3 Exit Mechanics and Risk Controls

To optimize capital velocity and protect the portfolio, the strategy enforces deterministic exit conditions anchored to raw physical prices rather than normalized signals.

- **Raw Spread Reversion:** Positions are liquidated when the raw, unadjusted price spread ($x_{A,B,t}$) crosses the rolling mean ($\mu_{raw,24h}$).

  This asymmetric metric application decouples signal conviction from PnL realization. While the dimensionless s-score is at entry to standardize statistical significance and filter market noise, portfolio PnL and exchange fees are realized exclusively in physical price space (basis points), not standard deviations. By anchoring the take-profit mechanism to the raw physical spread crossing the mean, the strategy guarantees cash-flow realization. It ensures the trade is closed exactly when the physical pricing dislocation has fully converged. In liquid and continuous markets, exiting based on the change in the normalized s-score is the optimal choice to guide fair price deviations; but for more volatile and fragmented markets, the more pragmatic choice is to prioritize PnL realizations from raw spreads.

- **Funding Epoch Blackouts:** Holding positions across asynchronous funding settlements requires additional capital for funding settlements. Positions are forcefully evacuated one minute prior to the nearest funding snapshot of either exchange, ensuring the strategy captures pure stochastic reversion without funding-rate noise.

- **Temporal Timeouts over Magnitude-Based Stop-Losses:** Traditional static stop-losses (e.g., exiting at a fixed $-X\%$ drawdown) were rigorously audited and deliberately excluded in favor of a hard 300-minute temporal timeout. In statistical arbitrage, adverse price excursions theoretically increase a trade's expected value, provided the equilibrium relationship remains intact. Furthermore, fragmented cryptocurrency derivative markets are uniquely susceptible to "liquidation cascades"—extreme, instantaneous wicks that resolve within seconds. A static price-based stop-loss may act as a forced liquidation trigger in these regimes and cause the strategy exits at maximum adverse liquidity just before reversion. To circumvent this, risk management is delegated to the dimension of time. Empirical observation indicates that true microstructure dislocations resolve well within a 5-hour window; a spread diverging beyond 300 minutes signals a fundamental structural break rather than a transient anomaly. By pairing this temporal timeout with the pre-trade *Illiquidity Entry Guard*, the architecture protects capital from prolonged directional exposure while remaining entirely immune to high-frequency data noise and false-positive liquidation wicks.

3 Experimental Design and Validation

3.1 Dataset Partitioning: In-Sample and Out-of-Sample

To evaluate the robustness of the mean-reversion strategy and mitigate the risk of over-fitting to specific market regimes, the dataset is partitioned into non-overlapping training and validation windows.

- **In-Sample Calibration (Mar 01, 2025 – Dec 31, 2025):** This twelve-month period serves as the training set. It is utilized for validating the OU model parameters, the risk management rules and the optimization of entry/exit s-score thresholds.
- **Out-of-Sample Validation (Jan 01, 2026 – Feb 28, 2026):** This period serves as an unseen test set. All model parameters are held static at their 2025 values to simulate a live deployment environment and assess the persistence of the identified arbitrage alpha.

3.2 Capital Allocation Set-up

To evaluate the performance of our strategy, we employ a pair-wise trading framework where each asset pair acts as an independent execution instance. In this setting, the strategy logic—governed by the same signal generator and risk engine—is applied simultaneously to every unique combination of venues and assets. This allows us to observe how the strategy behaves across different microstructure environments without the interference of inter-pair correlations or shared execution constraints.

For the purpose of this analysis, we assume an environment where each pair operates with an identical capital base. We do not impose capital constraints, rebalancing rules, or margin limitations between pairs. By treating each pair as an isolated unit, we can clearly identify which venue combinations provide reliable alpha and which environments introduce structural toxicity, effectively separating signal quality from portfolio-level capital management.

Performance is assessed through both independent and aggregate metrics. We first evaluate the outcomes for each pair individually to isolate venue-specific behaviors, such as latency gaps or data feed reliability. Following this, we compute an aggregate portfolio result by summing the returns of all trades across all pairs. This overall view allows us to calculate global performance metrics, such as the Sharpe ratio and cumulative PnL, providing a comprehensive assessment of the strategy's viability when deployed across the entire asset universe.

3.3 Asset Universe and Venue Selection

The asset universe is deliberately constructed to encompass a broad spectrum of market microstructures, liquidity regimes, and retail sentiment profiles. This structural diversity is essential for empirically testing the robustness of the execution architecture across varying states of market efficiency. All instruments evaluated within this study are linear perpetual futures denominated against USDT.

The research portfolio consists of four selected assets, each representing a distinct market regime:

- **Bitcoin (BTC) – The Institutional Benchmark:** Represents the highest liquidity tier. Its inclusion tests the strategy's viability and fee-sensitivity in efficient, institutionally dominated order books where arbitrage margins are heavily compressed.

- bfAvalanche (AVAX) – The Established Alt-L1: Represents mid-to-high capitalization Alternative Layer-1 networks. It offers a balance of deep underlying liquidity and retail-driven volatility, serving as the baseline for consistent, risk-adjusted alpha capture.
- **Berachain (BERA) – The High-Beta Alpha:** Represents emerging, narrative-driven assets with intense retail participation. It provides the high-frequency volatility and wider natural spreads required to test unconstrained mean-reversion profitability.
- **Kaito (KAITO) – The Toxicity Stress-Test:** Represents lower-liquidity, highly fragmented tokens. Characterized by frequent order book flatlining and severe execution friction, it serves to empirically validate the necessity and protective power of using stricter entry conditions.

To capture pricing dislocations across these distinct regimes, the cross-venue arbitrage matrix incorporates data from six major derivative exchanges: Binance, Bybit, OKX, Gate.io, Hyperliquid, and KuCoin. *Note:* The BTC matrix is constructed using only five venues, as KuCoin was excluded for the BTC analysis due to missing historical derivative ticker data during the procurement phase.

4 Data

4.1 Data Description and Sourcing

The empirical analysis is conducted using a high-frequency, multi-venue dataset spanning the evaluated cryptocurrency perpetual futures. To accurately reconstruct the physical trading environment and test the execution architecture, the dataset is synthesized from two primary sources:

1. **High-Frequency Tick Data (Tardis.dev):** Comprehensive derivative tick data, including funding rates, last traded prices and historical index prices, is sourced from Tardis.dev. This provides the necessary macro-market context and exact funding epoch timestamps required for the strategy's baseline calculations.
2. **Historical Level-1 Order Book Data (BBO):** To model accurate execution friction and calculate the real-time physical mid-spread ($x_{A,B,t}$), the engine utilizes a minute-level Best Bid/Offer (BBO) dataset. This localized feed captures the exact top-of-book executable liquidity across the six selected venues, ensuring that the backtest reflects actual market depth rather than theoretical, frictionless prices.

4.2 Data Integrity and Lookahead Bias Mitigation

The most prevalent point of failure in statistical arbitrage backtesting is the inadvertent introduction of lookahead bias. To guarantee that the execution engine operates in a causal, out-of-sample reality, two structural data-processing constraints are enforced at the pipeline level:

- **Causal Time-Series Alignment (Right-Labeled Bucketing):** During the aggregation of high-frequency tick data into minute-level actionable arrays, the time-series resampling is bound to right-labeled temporal bucketing. Any micro-structural data arriving at 10:00:30 is timestamped into the 10:01:00 execution bucket. This mandatory lag physically prevents the model from "seeing" intra-minute price movements, forcing the execution engine to make trading decisions based solely on the fully realized order-book state of the preceding minute.
- **Ex-Ante Feature Engineering:** All rolling statistical parameters—most notably the 1440-minute rolling mean (μ_{eq}), standard deviation (σ), and median liquidity toll (τ_{liq})—are shifted by $t-1$ relative to the execution timestamp. This $t-1$ barrier guarantees that the entry and exit gates are evaluated using only historical data, completely isolating the signal generation from the current-period execution price.

4.3 Data Cleaning and Anomaly Handling

Given the fragmented and volatile nature of cryptocurrency derivative exchanges, raw BBO and tick data are susceptible to websocket drops and missing periods. To preserve the integrity of the rolling statistics without introducing artificial price movements, missing physical spread data is forward-filled only up to a temporal limit. If an exchange experiences a prolonged data stall (e.g., a localized outage), the affected asset is flagged as "stale," triggering the engine's *Critical Force-Close Override*. This cleaning protocol ensures the backtest does not hold "zombie trades" through periods of unverified pricing data, maintaining exact parity with live execution safeguards.

Also, to secure adherence to the theoretical constraints of the model, particularly the 300-minute maximum holding parameter (τ_{max}), the execution engine employs a *Terminal Lookback Imputation* protocol. During empirical testing, standard forward-filling (`ffill`) methodologies were found to paralyze the engine by artificially projecting stale, wide spreads across API data gaps, effectively locking out valid execution signals. By deferring organic mean-reversion exits to localized liquidity events, but imputing the last-known valid BBO state for hard timeouts and funding blackout evacuations, the model maintains execution continuity without generating lookahead bias or phantom alpha.

4.4 Statistical Analysis

4.4.1 Stylized Facts across the Asset Universe

To establish a baseline for strategy performance, we examine the trading and cointegration dynamics across the identified asset universe. Given the extensive number of cross-venue permutations, we present a representative cross-section for each of the four tokens. For each asset, we isolate two distinct microstructural regimes: the *Institutional Anchor* (representing the deepest, lowest-variance venues dominated by consensus pricing, such as Binance/Bybit) and the *Alternative Fragmented Tail* (representing venues characterized by idiosyncratic retail order flow, decentralized execution dynamics, and elevated basis volatility, typically involving Gate or Hyperliquid).

Table 1: Stylized Facts: Representative Cross-Section of the Asset Universe

Asset	Representative Pair (Regime)	Mean Spread (Basis Pts)	SD of Spread (Basis Pts)	Req. Z-Score	Roll Stat Windows (%)	Active Days
BTC	BINA/BYBI (Anchor)	-1.17	2.07	4.82	98.5%	137.3
	GATE/HYPE (Tail)	-6.33	7.41	1.35	91.6%	239.2
AVAX	BINA/BYBI (Anchor)	0.14	2.46	4.06	100.0%	137.4
	GATE/HYPE (Tail)	-9.05	29.16	0.34	97.1%	239.8
BERA	BINA/BYBI (Anchor)	-1.79	5.97	1.67	100%	137.4
	GATE/HYPE (Tail)	-0.51	55.72	0.18	98.3%	239.6
KAITO	BINA/BYBI (Anchor)	-5.96	15.39	0.65	98.5%	136.3
	GATE/HYPE (Tail)	9.41	23.56	0.42	97.1%	239.6

Notes: **Req. Z-Score:** Theoretical standard deviations required to overcome a 10.00 bps Minimum Arbitrage Band (round-trip taker fees + 2.0 bps impact). **Roll Stat Windows (%):** The percentage of non-overlapping 24-hour rolling windows that successfully reject the null hypothesis of a unit root (ADF p-value < 0.05). This metric quantifies the continuous, day-to-day reliability of the mean-reverting relationship. **Active Days:** Total days of concurrent data availability. *BERA rolling stationarity inferred from global continuous cointegration checks.

The empirical results reveal several structural dynamics essential for strategy design:

1. **Global Cointegration vs. Continuous Tradability:** A prerequisite for statistical arbitrage is robust mean-reversion. First, global Augmented Dickey-Fuller (ADF) tests yielded p-values of 0.00 uniformly across the matrix, firmly rejecting the unit root hypothesis. This confirms that over the full sample, these spreads do not diverge infinitely and are structurally cointegrated. However, full-sample cointegration does not guarantee high-frequency tradability, as a pair could theoretically diverge for days at a time. To address this, the *Roll Stat Windows (%)* acts as a proxy for intraday safety. Across all ten pairs, the rolling stationarity remains exceptionally

high (ranging from 91.6% to 100.0%). This proves that the mean-reverting property is not just a macro-phenomenon, but is active and resolvable almost every single day, minimizing the risk of multi-day capital lockups.

2. **Structural Non-Zero Means and Index Disparities:** Across all pairs, the empirical mean of the premium-adjusted spread (μ) rarely anchors precisely at zero. Because the data pipeline evaluates spreads that have been neutralized back to index space, this constant deviation is not a byproduct of directional funding rates, but rather venue-specific disparities. This validates the architectural decision to anchor the trading engine to a rolling moving average rather than an absolute zero-price baseline, preventing the model from executing "ghost-arbitrage" against structural differences.

3. **Volatility Regimes and Profitability Hurdles:** The data categorizes the asset universe into two distinct operational regimes based on the standard deviation magnitude required to overcome the rule-fo-thumb 10-basis-point execution cost hurdle:

   - *Low-Volatility Regime (e.g., BTC Matrix):* These pairs exhibit severe spread compression (~2.0 bps variance). Then clearing fees requires extreme price divergence—roughly 4.8 standard deviations. This indicates that profitable arbitrage opportunities in institutional assets are statistically rare and sensitive to execution friction.
   - *High-Volatility Regime (e.g., BERA, KAITO Tails):* These fragmented pairs demonstrate massive baseline volatility ($>$ 20.0 bps). As a result, the required Z-score drops below 1.0σ. This regime may provide more frequent entry signals given the same cost hurdle, but in practice the spread cost may also be wider for altcoins and memecoins.

4. **Fat Tails (Excess Kurtosis):** The significant kurtosis universally observed across the asset matrices confirms that cryptocurrency spread distributions exhibit "fatter tails" than normal Gaussian distributions. This implies that extreme macro-dislocations ($> 3.0\sigma$) occur far more frequently than a standard normal model would predict, providing the execution engine with dense, localized clusters of high-conviction entry opportunities during market shocks.

5. **Data Asymmetry and Active History Limitations:** The total number of active trading days varies significantly, primarily due to missing historical data on specific venues. This data asymmetry necessitates our use of localized rolling 24-hour windows for parameter updates, ensuring that pairs are evaluated purely on their immediate structural merits.

4.4.2 Ornstein-Uhlenbeck (OU) Validation and Predictive Utility

To validate the statistical fit and predictive power of the mean-reverting processes, we model the pair spreads using a discrete-time approximation of the Ornstein-Uhlenbeck (OU) process. The OU model governs continuous-time mean reversion subject to random Brownian shocks, making it optimal for statistical arbitrage. We evaluate both the structural accuracy of the model (Table 2) and its ability to forecast physical reversion times in live order books (Table 3).

The merged validation framework confirms the strategy's mathematical viability through three insights:

1. **High-Precision Anchoring:** The *Mean Delta (Δ)* acts as a structural safety check. Across the anchor regimes, the theoretical OU equilibrium deviates from physical reality by less than 0.03 basis points. Even in extreme high-beta tails (BERA), the deviation maxes out at ~1.2 bps. This confirms the model accurately maps to physical pricing realities without generating theoretically misaligned equilibria.

2. **Predicting Physical Reversion:** Table 3 proves the OU model is effective at forecasting actual order-book behavior. Assets with faster AR(1) decay parameters (the altcoin matrices) inherently resolve their physical dislocations significantly faster than BTC. Crucially, the Spearman Rank Correlation between theoretical half-life and empirical 1σ TTR is exceptionally strong across all fragmented assets ($>$ 0.81). This confirms that the model's theoretical decay speed is a reliable proxy for live execution holding periods.

Table 2: OU Model Diagnostics: Precision of Mean Anchoring

Asset	Representative Pair (Regime)	AR(1) Beta (β)	Mean Delta (Δ bps)	Ljung-Box p-value
BTC	BINA/BYBI (Anchor)	0.9547	0.0003	0.000
	GATE/HYPE (Tail)	0.9926	0.0300	0.000
AVAX	BINA/BYBI (Anchor)	0.9092	0.0002	0.000
	GATE/HYPE (Tail)	0.9925	0.3363	0.000
BERA	BINA/BYBI (Anchor)	0.9275	0.0011	0.000
	GATE/HYPE (Tail)	0.9957	1.2623	0.000
KAITO	BINA/BYBI (Anchor)	0.9697	0.0268	0.000
	GATE/HYPE (Tail)	0.9921	0.1734	0.000

Notes: **Mean Delta (Δ):** Absolute difference between the theoretical model equilibrium (μ) and the empirical physical mean ($\bar{x}$). Values approaching zero confirm the model does not generate "ghost" equilibria.

Table 3: Predictive Utility: Theoretical vs. Empirical Anchored Time-to-Reversion (TTR)

Asset Universe (Cross-Section)	Average Theo. Half-Life (mins)	Avg. Empirical TTR at 1σ (mins)	Avg. Empirical TTR at 3σ (mins)	Spearman Rank Corr.
BTC Matrix	9.77	73.15	140.55	0.527
AVAX Matrix	4.66	28.33	48.63	0.864
BERA Matrix	5.03	23.60	44.27	0.835
KAITO Matrix	4.99	22.10	29.93	0.818

Notes: **Anchored TTR:** Realized time required for the physical spread to revert to its exact 24-hour causal moving average ($\mu_{eq,t}$) following a threshold breach, evaluated ex-ante to prevent lookahead bias.

3. **Residual Autocorrelation and Execution Design:** The Ljung-Box test (Table 2) yields p-values of 0.000 universally across the matrix. While the AR(1) structure captures the primary mean-reversion, the presence of residual autocorrelation indicates that high-frequency microstructure friction contains additional structural information. This limitation directly justifies our pragmatic architectural decision to rely on raw physical spreads—rather than pure statistical metric changes—for triggering live take-profit liquidations.

5 Core Results and Discussion

To establish an empirical performance baseline, the strategy was evaluated across all four assets utilizing a signal entry threshold of $|4.0\sigma|$ in an unconstrained execution environment (i.e., operating without an illiquidity spread guard). The aggregate global metrics and pair-level distributions are summarized in Panel A of Table 4.

5.1 Initial Alpha Capture and Structural Constraints

In the absence of an illiquidity guard, the portfolio exhibited robust systemic alpha in BTC, AVAX, and BERA. BTC and AVAX achieved unannualized cumulative returns of 432.66 bps and 6128.88 bps, translating to competitive Active Sharpe ratios of 5.08 and 2.83, respectively. This efficiency is further corroborated by the Gain-to-Pain (G2P) ratios; AVAX demonstrated exceptional absolute risk efficiency with a G2P of 4.30, indicating that gross positive returns overwhelmingly dwarfed negative bleed. BTC and BERA similarly maintained strong G2P ratios of 2.53 and 2.20, successfully capturing meaningful dislocation alpha in both deep and mid-tier liquidity assets.

Table 4: Portfolio Performance Across Unconstrained and Guarded Execution Regimes

	Overall Metrics							**Pair-Level Performance**		
Asset	**Trades**	**Win %**	**Cum PnL**	**Act SR**	**G2P**	**Max Loss**	**MDD**	**Winning Pairs**	**Best Performing**	**Worst Performing**
Panel A: Baseline Unconstrained Model (Max Entry Spread = ∞)										
BTC	117	44.4%	432.66 bps	5.08	2.53	-19.55 bps	57.92 bps	4 / 6	BINF/HYPE (+1.88%)	BINF/OKEX (-0.08%)
AVAX	132	68.2%	6128.88 bps	2.83	4.30	-893.46 bps	141.47 bps	8 / 11	BINF/OKEX (+39.65%)	GATE/HYPE (-8.06%)
BERA	127	59.1%	1498.57 bps	4.73	2.20	-86.31 bps	473.77 bps	6 / 12	GATE/HYPE (+8.68%)	BYDE/KUSW (-1.29%)
KAITO	307	53.1%	-1896.78 bps	-3.32	0.64	-444.61 bps	1896.78 bps	4 / 12	BINF/HYPE (+2.50%)	BYDE/KUSW (-7.00%)
Panel B: Microstructure Guarded Model (Max Entry Spread = 10.0 bps)										
BTC	117	44.4%	415.58 bps	5.06	2.47	-19.55 bps	57.92 bps	4 / 6	BINF/HYPE (+1.71%)	BINF/OKEX (-0.08%)
AVAX	118	66.9%	1375.57 bps	3.71	3.88	-38.95 bps	141.47 bps	8 / 10	BINF/OKEX (+11.26%)	BINF/KUSW (-0.12%)
BERA	71	57.8%	690.56 bps	3.88	2.32	-51.89 bps	169.84 bps	6 / 11	BINF/HYPE (+3.11%)	BYDE/OKEX (-0.81%)
KAITO	201	58.7%	-120.48 bps	-0.31	0.94	-377.85 bps	951.71 bps	6 / 12	GATE/HYPE (+2.88%)	BINF/KUSW (-4.73%)

Notes on Metric Definitions: Cum PnL is the unscaled sum of basis points generated across the backtest period. **Act SR** (Active Sharpe) is the annualized Sharpe ratio calculated from daily aggregated PnL, excluding days with zero capital deployment. **G2P** (Gain-to-Pain Ratio) is the sum of all positive trade returns divided by the absolute sum of all negative trade returns, measuring absolute risk efficiency. **Max Loss** identifies the single most toxic execution at the individual trade level. **MDD** (Maximum Drawdown) tracks the peak-to-trough decline of the *daily aggregated* equity curve. As a result, a single intraday *Max Loss* may exceed the *MDD* if offset by concurrent winning trades (intra-portfolio netting). Conversely, if an asset (e.g., KAITO Panel A) remains entirely in a state of decay without achieving a new equity peak, the MDD mathematically converges to the absolute Cumulative PnL. **Winning Pairs** measures the frequency of venue-combinations yielding positive net returns, while **Best/Worst Performing** highlight the specific venue pairs driving the extreme bounds of the asset's total return profile.

Table 5: Trade Termination Frequency and Moving Average Convergence (Baseline Unconstrained Model)

Asset	**Total Trades**	**MA Reversion Exits**	**Funding Epoch Exits**	**Timeout Exits**	**Overall MA Hit %**
BTC	117	45	71	1	38.46%
AVAX	132	77	53	2	58.33%
BERA	127	67	60	0	52.76%
KAITO	307	132	175	0	43.00%

Notes on Routing Categorization: Exit frequencies sum both standard and algorithmically delayed/imputed closure mechanics. **MA Reversion Exits** includes trades closed natively upon physical mean crossing, as well as trades where closure was queued during a data gap and executed organically upon liquidity restoration. **Funding Epoch Exits** captures forced evacuations driven by end- of-epoch triggers, utilizing lookback imputation where necessary. **Timeout Exits** aggregates natural 300-minute hold limit breaches. The **Overall MA Hit %** serves as a proxy for *Signal Purity*, indicating the percentage of total capital deployments that successfully reached theoretical equilibrium without requiring an artificial microstructure override.

However, a granular review of the termination mechanics (Table 5) reveals structural idiosyncrasies that artificially cap theoretical profitability. Despite achieving an Active Sharpe exceeding 5.0, BTC's win rate (44.4%) and overall moving-average reversion hit rate (38.46%) are notably lower than those of AVAX (58.33%) and BERA (52.76%). This discrepancy is an artifact of the strategy's risk-routing matrix rather than signal decay. As observed, 71 of BTC's 117 trades were forcefully evacuated due to Funding Epoch Exits. Because a vast majority of these executions involve Hyperliquid—which enforces a stringent 1-hour funding interval—the empirical Time-to-Reversion (TTR) is frequently truncated. With the average hold time sitting at approximately 31.5 minutes, the engine is forced to liquidate positions prematurely to avoid structural funding penalties, often terminating the trade before the spread has sufficient time to organically revert to its moving-average mean.

5.2 Microstructure Toxicity and Tail-Risk Identification

While the unconstrained regime captures significant nominal returns in stable assets, it simultaneously exposes the portfolio to severe structural vulnerabilities when applied to fragmented or long-tail assets. This is most evident in KAITO, which, despite achieving a theoretically positive win rate of 53.1%, generated a Cumulative PnL of -1896.78 bps and a Maximum Drawdown equal to its total loss. This systemic decay is formalized by KAITO's G2P ratio of 0.64, confirming that the magnitude of losing executions vastly outpaces any accumulated winning trades.

A granular review of KAITO's termination profile highlights that a 175 of its 307 trades ended via forced Funding Epoch Exits rather than organic MA reversion. Crucially, this high frequency of funding liquidations must be distinguished from the phenomenon observed in BTC. As demonstrated in Table 3, BTC possesses a naturally prolonged empirical Time-to-Reversion (TTR of 140.55 minutes at 3σ), causing genuine trades to be prematurely truncated by funding clocks. KAITO, conversely, exhibits a rapid empirical TTR of just 29.93 minutes.

The fact that the majority of KAITO's unconstrained executions languished in the portfolio until a forced funding exit—vastly exceeding its natural 30-minute reversion profile—reveals a microstructure failure. These toxic entries did not "structurally fail to revert"; rather, they were *phantom spreads* triggered by transient liquidity vacuums and order book flatlines. Because these executions were initiated on data artifacts rather than genuine stochastic deviations, they lacked a true mean-reverting catalyst. As a result, capital was trapped in deteriorating, non-reverting noise until the exchange epoch forced a realization of the accumulated loss. The prevalence of these phantom entries firmly establishes the necessity of a quantitative filter—specifically, a microstructure spread guard to verify execution validity and truncate tail risk during liquidity disruptions.

5.3 Efficacy of the Microstructure Guard and Regime Heterogeneity

The application of a static 10.0 bps illiquidity guard (Panel B) reveals a stark risk-reward trade-off that varies significantly across the asset cross-section. For the top three liquidity assets (BTC, AVAX, and BERA), the rigid microstructural constraint successfully achieved its primary objective: the truncation of tail risk. Most notably, AVAX's single-trade Max Loss was curtailed from -893.46 bps to a manageable -38.95 bps, while BERA's Maximum Drawdown was compressed from 473.77 bps to 169.84 bps.

However, this systemic protection exacted a heavy toll on baseline capacity. Both AVAX and BERA experienced severe contractions in Cumulative PnL, sacrificing over 4753 bps and 808 bps of nominal yield, respectively. This volume contraction indicates that the 10.0 bps guard is overly punitive for deeper liquidity regimes, prematurely blocking genuine high-volatility statistical arbitrage opportunities under the guise of toxicity.

In contrast, the exact same 10.0 bps guard proved efficacious for KAITO. Dominated by microstructure noise, KAITO required this intervention to survive. By filtering out non-reverting phantom spreads and preventing 106 toxic executions, the guard slashed KAITO's cumulative loss from an unmanageable -1896.78 bps down to a negligible -120.48 bps.

This heterogeneous response highlights a fundamental limitation of utilizing a uniform risk parameter across fragmented crypto markets. Deep-liquidity regimes are structurally penalized by tight constraints that choke alpha generation, whereas high-toxicity regimes require aggressive filtering to prevent capital decay. This divergence provides empirical evidence that varying market microstructures demand asset-specific execution parameters to achieve the optimal trade-off between opportunity capture and adverse selection. To maximize Gain-to-Pain efficiency, it is necessary to map the strategy's performance across a multi-dimensional parameter space. Section 6 details a systematic sensitivity analysis—varying both the signal conviction threshold (σ) and the illiquidity spread guard—to derive regime-specific optimal calibrations.

5.4 Robustness Check: Normalized OU Framework vs. Naive Z-Score

To rigorously validate our hypothesis that using normalized price with s-scores can help improve filtering falsely positive signals, we conduct a structural robustness check against a simpler statistical baseline. We do a head-to-head comparison on the unconstrained BTC strategy: once utilizing the naive rolling Z-score, and once utilizing the funding-normalized OU S-Score. The aggregate performance metrics, presented in Table 6, definitively prove the superiority of the normalized OU framework across all key performance indicators.

Table 6: Robustness Check: Naive Raw Z-Score vs. Normalized OU S-Score (Unconstrained BTC)

Model	**Sigma (σ)**	**Trades**	**Win %**	**Cum PnL**	**Act SR**	**Trd SR**	**MDD**	**G2P**
Naive Raw Z-Score	3.0σ	333	23.1%	-440.50 bps	-2.51	-0.10	741.69 bps	0.64
Naive Raw Z-Score	3.5σ	247	29.2%	-125.75 bps	-1.02	-0.03	438.33 bps	0.85
Naive Raw Z-Score	4.0σ	181	37.6%	265.95 bps	2.37	0.09	139.57 bps	1.53
Normalized OU S-Score	3.0σ	225	27.6%	-104.39 bps	-0.81	-0.03	408.38 bps	0.87
Normalized OU S-Score	3.5σ	162	36.4%	247.17 bps	2.40	0.08	133.09 bps	1.52
Normalized OU S-Score	4.0σ	117	44.4%	**432.66 bps**	**5.08**	**0.18**	**57.92 bps**	**2.53**

Notes: Act SR = Active Sharpe Ratio; Trd SR = Trade-Level Sharpe Ratio; MDD = Maximum Drawdown; G2P = Gain-to-Pain Ratio. Results simulated over the unconstrained BTC matrix.

The most immediate divergence between the two models is execution frequency. At the 4.0σ threshold, the naive model generated 181 trades, whereas the OU model executed only 117. This implies that 64 executions—representing over 35% of the naive model's volume—were toxic false positives which can be driven by deterministic funding drift rather than genuine stochastic dislocations. By filtering out these structural traps, the OU model's Win Rate climbs from 37.57% to 44.44%. The elimination of these falsely positive trades has a profound impact on the portfolio's risk-reward efficiency. Despite executing 35% fewer trades at the 4.0σ level, the normalized OU model captures substantially more absolute yield (432.66 bps vs. 265.95 bps). Crucially, the robustness check highlights the strategy's enhanced capacity for risk compression. By avoiding funding traps, the Maximum Drawdown (MDD) is slashed by nearly 60%, dropping from a vulnerable 139.57 bps under the naive model to a tight 57.92 bps under the OU model. Furthermore, the Gain-to-Pain (G2P) ratio expands from 1.53 to an elite 2.53.

6 Multi-Dimensional Parameter Sensitivity and Optimization

To systematically determine the boundary conditions of the strategy and isolate optimal execution regimes, we conducted a multi-dimensional parameter sensitivity analysis (Table 9). The evaluation grid toggled the signal conviction threshold ($\sigma \in \{2.0, 2.5, 3.0, 3.5, 4.0\}$) against varying levels of microstructural stringency (Unconstrained vs. Guarded at 5.0, 10.0, and 20.0 bps).

Observations on Signal Conviction (σ)

A persistent characteristic observed across all asset matrices is the monotonic relationship between signal conviction and portfolio expectancy. Relaxing the entry threshold from 4.0σ down to 2.0σ artificially hyper-inflates trade frequency—for instance, BTC capital deployments spike from 117 to 517 trades. However, this increased capacity systematically degrades both the Win Rate and the Active Sharpe ratio.

The degradation of performance at lower conviction thresholds (e.g., 2.0σ) also exposes a limitation in the interaction between the EV gate and the funding epoch blackout rule. While the EV Gate mandates a net-positive expected return prior to execution, its calculation implicitly assumes the spread will achieve full convergence to the moving average ($\mu_{raw,24h}$).In institutional assets like BTC, spread volatility is heavily compressed. At a 2.0σ entry, the gross physical spread barely exceeds the aggregate transaction cost hurdle (C_{trip}), resulting in a marginal expected value. When the strategy rule forcefully liquidates these positions prior to an epoch snapshot, the holding period is truncated before full mean-reversion can materialize. At a 4.0σ entry, the pricing dislocation is sufficiently wide that even a partial reversion yields enough spread compression to clear taker fees. This confirms that in low-variance regimes, high signal conviction is required to provide a sufficient margin of safety against time-forced liquidations.

Regime-Specific Guard Tolerances

The sensitivity grid empirically isolates the differing liquidity profiles of the evaluated assets. Institutional anchors like BTC demonstrate near-perfect invariance to tighter liquidity guards; the performance profile at a 10.0 bps guard is virtually identical to the unconstrained profile, confirming the depth of top-tier order books. Conversely, KAITO completely fails to generate a positive expectancy under unconstrained or wide-guard regimes. It only achieves a positive cumulative return (+166.55 bps) at the absolute intersection of maximum conviction (4.0σ) and maximum microstructural protection (5.0 bps guard).

Systematic Parameter Selection for Out-of-Sample (OOS) Validation

To eliminate selection bias for the subsequent 2026 Out-of-Sample (OOS) testing, we establish an optimization protocol. The objective function maximizes the *Active Sharpe Ratio* subject to two constraints: (1) Cumulative PnL must be positive, and (2) in the event of equivalent Sharpe ratios, the configuration with the widest permissible guard is selected to maximize theoretical capacity. Note that this calibration evaluates cases with bounded illiquidity guards only. Applying this protocol to the sensitivity matrix yields the following parameter choices (Table 7), which will serve as the execution parameters for the forward-testing phase:

Table 7: Systematic Optimal Parameter Calibration for OOS Testing

Asset	Regime Type	Optimal Guard	Optimal Sigma
BTC	Deep Liquidity	10.0 bps	4.0σ
AVAX	High Conviction	20.0 bps	4.0σ
BERA	Alpha-Dense	10.0 bps	4.0σ
KAITO	High Toxicity	5.0 bps	4.0σ

7 Out-of-Sample Forward Validation

To rigorously validate the predictive utility of the statistical arbitrage engine and the robustness of the microstructure guards, the strategy was subjected to an out-of-sample (OOS) forward test over a 2-month horizon in 2026. The execution engine was locked to the optimal regime-specific calibrations derived from the in-sample Pareto protocol (all signal thresholds fixed at 4.0σ, with illiquidity guards set to 10.0 bps for BTC/BERA, 20.0 bps for AVAX, and 5.0 bps for KAITO). The aggregate findings, detailed in Table 8, present a distinct profile characterized by exceptional execution precision bounded by severe capacity constraints.

The primary objective of the OOS evaluation was to confirm that the strategy successfully avoids in-sample curve fitting. The results definitively prove the out-of-sample persistence of the alpha signal. Across the evaluated regimes, the engine delivered exceptionally high accuracy and risk-adjusted efficiency. AVAX achieved a flawless 100% win rate across its deployments, recording an Active Sharpe ratio of 21.66.

BERA emerged as the strongest absolute performer, capturing 336.88 bps over the 2-month period with an impressive 78.26% win rate. Furthermore, BERA's Gain-to-Pain (G2P) ratio of 5.55 and a heavily truncated Maximum Drawdown of just 17.51 bps confirm that the 10.0 bps microstructure guard successfully transferred its protective capabilities to unseen data. Even in fragmented assets like KAITO, where the nominal win rate dropped to 33.3%, the structural logic of the engine prevailed: the magnitude of the single winning trade dwarfed the minor losses, yielding a positive G2P of 4.05 and a net positive cumulative return.

While the OOS performance exhibits robust risk efficiency, the combination of a 4.0σ entry threshold and tight microstructure guards effectively limit execution capacity. Over the 2-month forward-testing horizon, the engine executed only 12 trades for BTC, 5 trades for AVAX, and a mere 3 trades for KAITO. Absolute returns of 18.90 bps (BTC) or 21.00 bps (AVAX) over a 60-day window are insufficient to overcome base capital opportunity costs without the application of high leverage.

Table 8: Forward-Test Aggregate Metrics and Pair-Level Performance (2-Month Out-of-Sample)

	Overall OOS Metrics							**Pair-Level Performance**		
Asset (Config)	**Trades**	**Win %**	**Cum PnL**	**Act SR**	**G2P**	**Max Loss**	**MDD**	**Winning**	**Best Performing**	**Worst Performing**
BTC (10 bps)	12	58.3%	18.90 bps	7.34	2.82	-5.14 bps	5.65 bps	2 / 4	BINF/OKEX (+0.18%)	GATE/HYPE (-0.05%)
AVAX (20 bps)	5	100.0%	21.00 bps	21.66	N/A	+1.29 bps	0.00 bps	2 / 2	BINF/HYPE (+0.14%)	BINF/OKEX (+0.07%)
BERA (10 bps)	23	78.3%	336.88 bps	11.43	5.55	-20.22 bps	17.51 bps	4 / 4	BINF/OKEX (+1.92%)	BINF/HYPE (+0.14%)
KAITO (5 bps)	3	33.3%	36.03 bps	7.39	4.05	-7.06 bps	7.06 bps	1 / 1	BINF/OKEX (+0.36%)	N/A –

Notes on OOS Evaluation: Performance metrics reflect a 2-month out-of-sample forward test (2026 data). Assets were executed using their optimal regime configurations derived from the in-sample Pareto-Optimization protocol (all thresholds fixed at 4.0σ, with varying asset-specific microstructure guards). A G2P of N/A (AVAX) indicates an absence of realized negative trade returns during the OOS window. For KAITO, the strategy only executed on a single venue pair (BINF/OKEX).

Table 9: Strategy Sensitivity Analysis across Thresholds and Illiquidity Guards

Asset	Guard Type	Sigma	Trades	Win %	Cum PnL	Cal SR	Act SR	Trd SR	G2P	Max Loss	MDD	Avg Hold
BTC	Unconstrained	2.00	517	18.2	-1413.62	-4.58	-7.73	-0.34	0.34	-19.55	1447.10	27.8
		2.50	357	20.7	-625.57	-2.04	-3.56	-0.13	0.56	-19.55	878.71	27.9
		3.00	225	27.6	-104.39	-0.42	-0.81	-0.03	0.87	-19.55	408.38	28.8
		3.50	162	36.4	247.17	1.15	2.40	0.08	1.52	-19.55	133.09	30.5
		4.00	117	44.4	432.66	2.01	5.08	0.18	2.53	-19.55	57.92	31.5
	Guarded (5.0 bps)	2.00	516	18.0	-1369.66	-4.21	-7.03	-0.29	0.36	-19.55	1437.66	27.8
		2.50	357	20.7	-669.09	-2.35	-4.12	-0.16	0.53	-19.55	878.71	27.9
		3.00	225	27.6	-147.91	-0.65	-1.24	-0.05	0.82	-19.55	408.38	28.9
		3.50	162	36.4	203.66	1.02	2.13	0.08	1.43	-19.55	133.09	30.6
		4.00	116	43.1	358.91	1.83	4.60	0.17	2.26	-19.55	57.92	32.0
	Guarded (10.0 bps)	2.00	517	18.2	-1413.62	-4.58	-7.73	-0.34	0.34	-19.55	1447.10	27.8
		2.50	357	20.7	-642.65	-2.16	-3.77	-0.14	0.55	-19.55	878.71	27.9
		3.00	225	27.6	-121.47	-0.51	-0.97	-0.04	0.85	-19.55	408.38	28.9
		3.50	162	36.4	230.10	1.10	2.31	0.08	1.49	-19.55	133.09	30.6
		4.00	117	44.4	415.58	2.00	5.06	0.18	2.47	-19.55	57.92	31.6
	Guarded (20.0 bps)	2.00	517	18.2	-1413.62	-4.58	-7.73	-0.34	0.34	-19.55	1447.10	27.8
		2.50	357	20.7	-638.57	-2.13	-3.72	-0.14	0.55	-19.55	878.71	28.0
		3.00	225	27.6	-117.39	-0.49	-0.93	-0.03	0.86	-19.55	408.38	28.9
		3.50	162	36.4	234.18	1.12	2.33	0.08	1.50	-19.55	133.09	30.6
		4.00	117	44.4	419.66	2.01	5.06	0.18	2.49	-19.55	57.92	31.7
AVAX	Unconstrained	2.00	321	53.6	4797.52	1.15	2.06	0.06	2.93	-892.81	364.72	28.5
		2.50	239	55.7	4465.93	1.15	2.26	0.07	3.00	-892.81	263.57	29.2
		3.00	178	61.8	5446.29	1.17	2.56	0.09	3.73	-892.81	140.91	29.5
		3.50	152	66.5	5666.09	1.21	2.75	0.10	3.99	-892.81	140.91	31.3
		4.00	132	68.2	6128.88	1.19	2.83	0.12	4.30	-893.46	141.47	32.3
	Guarded (5.0 bps)	2.00	207	40.1	-275.67	-1.42	-2.83	-0.13	0.67	-33.91	321.94	30.1
		2.50	149	44.3	-180.49	-1.17	-2.52	-0.12	0.70	-33.91	236.05	30.9
		3.00	103	49.5	-82.68	-0.59	-1.41	-0.07	0.81	-33.91	140.23	31.5
		3.50	85	55.3	80.53	0.47	1.19	0.07	1.22	-33.91	117.72	33.4
		4.00	74	58.1	94.99	0.57	1.51	0.09	1.29	-33.91	121.88	36.2

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Asset	Guard Type	Sigma	Trades	Win %	Cum PnL	Cal SR	Act SR	Trd SR	G2P	Max Loss	MDD	Avg Hold
	Guarded (10.0 bps)	2.00	295	50.5	1039.12	1.14	2.08	0.06	1.94	-38.95	373.40	29.0
		2.50	218	52.8	1065.01	1.18	2.36	0.07	2.23	-38.95	263.57	29.2
		3.00	160	59.4	1206.93	1.35	3.01	0.09	2.96	-38.95	140.91	29.2
		3.50	135	64.4	1409.20	1.55	3.65	0.11	3.71	-38.95	140.91	31.0
		4.00	118	67.0	1375.57	1.52	3.71	0.12	3.88	-38.95	141.47	31.8
	Guarded (20.0 bps)	2.00	313	53.0	3883.06	1.16	2.06	0.09	4.47	-38.95	364.72	28.8
		2.50	232	55.2	3699.63	1.16	2.28	0.11	5.24	-38.95	263.57	29.4
		3.00	172	61.6	3818.49	1.20	2.63	0.13	7.10	-38.95	140.91	29.8
		3.50	146	66.4	3999.00	1.26	2.87	0.14	8.55	-38.95	140.91	31.7
		4.00	128	68.8	3965.32	1.25	2.98	0.15	9.14	-38.95	141.47	32.7
BERA	Unconstrained	2.00	213	57.8	1231.77	1.66	3.52	0.12	1.67	-86.31	478.72	34.2
		2.50	186	57.0	1195.16	1.63	3.61	0.13	1.73	-86.31	473.66	35.8
		3.00	167	56.3	1115.96	1.55	3.50	0.13	1.71	-86.31	473.66	34.3
		3.50	153	58.8	1535.74	1.79	4.28	0.17	2.08	-86.31	471.29	35.0
		4.00	127	59.1	1498.57	1.76	4.73	0.19	2.20	-86.31	473.77	36.2
	Guarded (5.0 bps)	2.00	53	32.1	-111.91	-0.66	-2.38	-0.13	0.70	-44.03	282.12	46.7
		2.50	42	38.1	-14.55	-0.10	-0.36	-0.02	0.95	-44.03	208.11	51.0
		3.00	35	40.0	2.82	0.02	0.07	0.00	1.01	-44.03	187.45	49.5
		3.50	30	43.3	32.70	0.22	0.92	0.05	1.16	-44.03	163.81	54.5
		4.00	22	50.0	103.89	0.72	3.72	0.21	1.75	-44.03	113.10	58.2
	Guarded (10.0 bps)	2.00	132	53.0	481.74	0.96	2.25	0.12	1.54	-57.76	282.04	36.1
		2.50	111	54.1	521.21	1.00	2.43	0.14	1.68	-57.76	277.42	38.7
		3.00	97	54.6	601.09	1.15	2.89	0.18	1.87	-57.76	204.89	38.1
		3.50	87	56.3	650.66	1.23	3.26	0.20	2.01	-57.76	176.17	39.0
		4.00	71	57.8	690.56	1.32	3.88	0.25	2.32	-51.89	169.84	39.9
	Guarded (20.0 bps)	2.00	196	56.6	521.40	0.84	1.77	0.09	1.33	-61.12	478.72	35.6
		2.50	170	56.5	603.19	0.94	2.09	0.12	1.43	-57.76	473.66	37.6
		3.00	153	56.2	547.27	0.88	1.98	0.11	1.41	-57.76	473.66	36.3
		3.50	140	57.9	673.75	1.07	2.53	0.15	1.55	-57.76	471.29	37.0
		4.00	114	57.9	636.58	1.03	2.74	0.16	1.60	-56.71	473.77	38.8

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Table 9 – continued from previous page

Asset	Guard Type	Sigma	Trades	Win %	Cum PnL	Cal SR	Act SR	Trd SR	G2P	Max Loss	MDD	Avg Hold
KAITO	Unconstrained	2.00	777	44.4	-5458.21	-2.44	-3.74	-0.16	0.51	-457.82	5481.03	48.1
		2.50	578	46.9	-4271.89	-2.26	-3.85	-0.15	0.53	-457.82	4309.06	50.4
		3.00	442	50.0	-3450.05	-2.09	-3.97	-0.15	0.55	-457.82	3450.05	52.3
		3.50	375	51.5	-2850.80	-1.94	-3.97	-0.14	0.58	-444.61	2850.80	53.4
		4.00	307	53.1	-1896.78	-1.57	-3.32	-0.12	0.64	-444.61	1896.78	53.8
	Guarded (5.0 bps)	2.00	373	46.4	-909.72	-1.70	-2.90	-0.09	0.69	-273.86	1083.09	43.1
		2.50	257	49.0	-823.96	-1.36	-2.62	-0.10	0.66	-273.86	927.81	45.8
		3.00	169	55.0	-145.66	-0.39	-0.85	-0.03	0.91	-161.58	586.00	47.6
		3.50	134	59.0	-14.96	-0.04	-0.10	-0.00	0.99	-182.72	575.42	46.8
		4.00	108	60.2	166.55	0.52	1.25	0.05	1.17	-166.35	304.97	50.3
	Guarded (10.0 bps)	2.00	589	46.7	-2011.77	-1.87	-2.94	-0.10	0.66	-377.85	2034.58	47.7
		2.50	415	49.9	-1490.28	-1.33	-2.32	-0.10	0.68	-377.85	1527.45	50.3
		3.00	299	54.2	-1301.18	-1.23	-2.43	-0.11	0.65	-377.85	1301.18	49.4
		3.50	244	58.6	-433.36	-0.45	-0.95	-0.04	0.84	-377.85	1023.62	50.8
		4.00	201	58.7	-120.48	-0.14	-0.31	-0.02	0.94	-377.85	951.71	48.8
	Guarded (20.0 bps)	2.00	728	44.9	-3806.65	-2.37	-3.63	-0.14	0.55	-444.61	3829.46	48.3
		2.50	538	47.6	-2854.67	-1.95	-3.33	-0.13	0.58	-444.61	2891.84	50.8
		3.00	404	50.5	-2525.62	-1.72	-3.24	-0.14	0.56	-444.61	2525.62	52.8
		3.50	333	52.6	-1896.79	-1.40	-2.86	-0.12	0.61	-444.61	1896.79	54.5
		4.00	270	53.7	-1382.85	-1.19	-2.55	-0.11	0.64	-444.61	1696.21	54.0

Notes on Metric Definitions: Win %: The percentage of total trades where Net PnL > 0. **Cum PnL**: The absolute sum of daily net basis point returns across the backtest period. **Cal SR** (Calendar Sharpe): Annualized Sharpe ratio calculated over all calendar days ($\sqrt{365} \times \frac{\mu_{cal}}{\sigma_{cal}}$), treating non-trading days as 0.0 return. **Act SR** (Active Sharpe): Annualized Sharpe ratio calculated on active trading days ($\sqrt{365} \times \frac{\mu_{act}}{\sigma_{act}}$). **Trd SR** (Trade Sharpe): Unannualized Sharpe ratio calculated directly from the distribution of individual trade net PnLs ($\frac{\mu_{trd}}{\sigma_{trd}}$). **G2P** (Gain-to-Pain): The absolute ratio of the sum of all gross winning trades to the sum of all gross losing trades ($\frac{\Sigma \text{Gains}}{|\Sigma \text{Losses}|}$). **Max Loss**: The minimum individual net PnL realized on a single executed trade. **MDD** (Maximum Drawdown): The maximum peak-to-trough decline measured from the daily aggregated cumulative PnL curve. **Avg Hold**: The arithmetic mean of position holding times in minutes.

8 Simplified Assumptions, Limitations and Potential Improvements

To establish a baseline framework for statistical arbitrage, this study employs several structural and operational simplifications. While these assumptions isolate the raw alpha of the theoretical model, translating this framework to a production-grade algorithmic trading desk requires addressing real-world exchange constraints and microstructure frictions.

Data Constraints and Asymmetric Histories

- **Limitation:** The empirical results are bounded by data availability and asymmetric historical coverage across venues. For instance, Best Bid/Offer (BBO) quote data for specific exchanges (e.g., Bybit) terminates prematurely in July 2025 within our sample, and certain historical derivative ticker data for BTC was unavailable during procurement.
- **Justification:** To prevent lookahead bias and artificial equity curves, the execution engine relies on concurrent, verifiable data. Periods with missing data act as operational blackouts, preventing trade execution.
- **Impact & Improvement:** The reported cumulative returns and total trade counts are systematically underestimated, representing a partial empirical picture rather than the strategy's true theoretical capacity. Future iterations can resolve this by procuring more Level-1 and Level-2 tick data feeds across all evaluated venues, enabling a more continuous assessment of profitability.

Data Granularity and Asynchronous Execution Architecture

- **Limitation:** The empirical backtest is constrained to 1-minute resolution data. The model assumes it can compute the live stochastic signal (S-score) and execute against the real-time Best Bid/Offer (BBO) with effectively zero latency. In reality, 1-minute data obfuscates sub-second micro-structural decay, and true zero-latency execution is physically impossible, exposing the strategy to queue-position disadvantages and intra-minute adverse selection on fast-decaying assets.
- **Justification:** To attain computational realism and eliminate lookahead bias, the model mimics a production-grade "Asynchronous Execution Architecture" (Slow Path / Fast Path separation). All heavy econometric parameter estimations—including the Ornstein-Uhlenbeck regression (β), equilibrium mean (μ), and equivalent volatility (σ_{eq})—are computed on the $t-1$ lagged window (the Slow Path). When the live BBO price arrives at time t, computing the actionable S-score is reduced to a simple, non-blocking $O(1)$ arithmetic operation ($S_t = \frac{P_t - \mu_{t-1}}{\sigma_{eq,t-1}}$). Because this "Fast Path" calculation executes in microseconds in production environments, assuming immediate execution against the 1-minute BBO is a defensible simplifying assumption for this frequency domain.
- **Improvement:** To accurately model execution slippage on fast-decaying anomalies, future backtesting engines must upgrade from 1-minute snapshots to continuous Level-2 or Level-3 tick data. Furthermore, a synthetic latency penalty (e.g., 20 to 50 milliseconds) should be introduced between the signal generation and the modeled fill timestamp to accurately simulate network routing limits and exchange matching-engine delays.

The "Early Kill" Problem from Funding Evacuation

- **Limitation:** To isolate the stochastic mean-reversion signal from deterministic cash flows, the current execution engine employs a "Funding Evacuation" heuristic. Active positions are forcefully liquidated immediately prior to an exchange funding snapshot. While this acts as a hard safety guard against unknown funding liabilities, it introduces the "Early Kill" problem. Forcefully truncating trades limits the hold time, often prematurely terminating positions that were on a successful trajectory toward the target MA. We can observe that BTC pairs' low win rates and weak performances at lower sigmas can be particularly related to early closes of positions.

- **Justification:** In initial baseline testing, deterministic funding penalties on unaligned legs can severely distort the profitability of the stochastic spread capture. Evacuating before the epoch ensures the portfolio's Expected Value (EV) is generated from order-book liquidity shocks rather than cash-flow accounting anomalies.
- **Improvement & Extension:** Future iterations of this architecture should test a continuous "Hold-Through" execution regime. Holding a position across a funding epoch is not a negative EV event; the strategy may be positioned to receive net funding payments. Furthermore, post-epoch spread convergence frequently exhibits high velocity as arbitrageurs unwind snapshot positions. By integrating continuous funding rate data into the position's running PnL, the engine could dynamically evaluate whether the expected probability of an MA convergence outweighs the immediate cash cost of crossing the funding boundary. This would likely rescue a significant percentage of "early kill" trades and potentially increase total absolute yield on assets like BTC.

Execution Pricing and Market Impact Modeling

- **Limitation:** The current engine assumes trades execute entirely at the Best Bid/Offer (BBO) quotes. While this is accurate for minimal lot sizes, larger institutional capital deployments would inevitably consume the top of the order book, incurring slippage and market impact costs that vary drastically by asset depth.
- **Justification:** The model accounts for this baseline friction by utilizing an aggressive "Taker-Taker" fee assumption alongside a pre-trade Cost-Adjusted Expected Value (EV) Gate. Furthermore, the Illiquidity Entry Guard acts as an effective proxy to reject completely hollow books and prevent executions during severe liquidity vacuums.
- **Improvement:** To accurately model scalability, the execution simulator must integrate full Level-2 (L2) order book depth data (e.g., the top 25 bid-ask tiers). This would allow the engine to calculate a volume-weighted average price (VWAP) for every execution, explicitly modeling the slippage cost function against the specific market depth of the asset.

Capital Constraints

- **Limitation:** The strategy evaluates each pair as an independent execution instance operating with an identical, unconstrained capital base. In reality, cross-exchange perpetual arbitrage requires distinct capital and margin pools siloed across multiple exchanges.
- **Justification:** Evaluating pairs independently without margin constraints cleanly isolates signal quality and venue-specific behaviors from portfolio-level capital management limitations. By treating each pair as an independent, fully-funded unit, we can accurately measure the pure predictive utility of the signal without obfuscating the results through capital rationing.
- **Improvement:** The framework should be extended to simulate a unified, fixed-margin pool. In this environment, capital is finite and incurs an opportunity cost. To optimize capital efficiency across concurrent signals, heterogeneous weighting methodologies such as the Kelly Criterion could be deployed to dynamically allocate scarce capital based on real-time spread volatility and win probabilities.

Leverage and Liquidation Risks

- **Limitation:** The current results are presented on a purely unlevered (1x nominal) basis. As demonstrated in the Out-of-Sample results, deep-liquidity anchors (like BTC) generate accurate, low-drawdown signals, but the minimal nominal spread capture results in low absolute cumulative returns.
- **Justification:** Presenting unlevered returns makes the performance metrics (such as the Sharpe ratio and Win Rate) reflect the predictive utility of the statistical model, preventing leverage multipliers from obfuscating the underlying alpha generation.

- **Improvement:** Given the strategy's property of low-frequency signals and exceptionally low system drawdowns on institutional assets, future models should integrate dynamic leverage tiers. Applying leverage (e.g., 5x to 10x) would amplify the nominal yield of liquid assets to institutional requirements. However, this must be paired with rigorous stress-testing to model the amplified probability of exchange liquidations, particularly during the severe "liquidation cascades" uniquely prevalent in cryptocurrency markets.

9 Conclusion

This study validates the structural viability of systematic cross-exchange perpetual arbitrage, demonstrating that transient pricing dislocations can be profitably exploited when rigorous quantitative controls are applied. The empirical findings underscore four conclusions regarding the design of cryptocurrency arbitrage systems.

First, the research reconfirms the necessity of tackling the funding effect; raw empirical spreads are frequently illusory, and stripping funding rate mechanics via price normalization is required to reveal true arbitrage opportunities. Second, the deployment of the OU driven s-score proved effective. By standardizing deviations, the s-score provided a robust framework for identifying high-conviction mean-reverting entry signals across disparate market venues.

Third, the integration of the Bid-Ask Illiquidity Guard successfully solved the problem of phantom liquidity. By actively filtering out toxic microstructure flow and dynamically adjusting to bid-ask widening, the system effectively insulated the portfolio from severe execution slippage that would otherwise destroy theoretical margins. Finally, the success and persistence of the results during the Out-of-Sample (OOS) forward validation confirm that the strategy's alpha is structurally genuine rather than an artifact of in-sample curve fitting. Ultimately, when fortified by liquidity guards and execution constraints, cross-exchange perpetual arbitrage remains a resilient mechanism for alpha generation.

Appendix A: Institutional Transaction Cost Assumptions

To accurately simulate a production-grade high-frequency trading environment, the backtesting engine assumes institutional-tier transaction costs. Retail fee schedules are not representative of the structural execution costs faced by algorithmic market participants in cross-exchange arbitrage.

Table 10 outlines the top-tier VIP fee schedules for the perpetual futures markets utilized in the empirical analysis. All fees are expressed in basis points (bps). No rebates are assumed here.

Table 10: Highest VIP Tier Execution Fees by Exchange

Exchange Venue	**Assumed Tier**	**Maker Fee (bps)**	**Taker Fee (bps)**
Binance	VIP 9	0.00	1.70
Bybit	Pro 5 / Supreme VIP	1.00/0.00	3.20/3.00
OKX	VIP 9	0.00	1.75
KuCoin	VIP 12	0.00	2.50
Gate.io	VIP 16	0.00	2.00
Hyperliquid	Diamond	0.00	1.44

References

[1] Marco Avellaneda and John Lee. Statistical arbitrage in the us equities market. *Quantitative Finance*, 10(7):761–782, 2010.